

A Review of AI-and-Edge-Driven IoT Platforms for Smart Traffic and Parking Management in Evolving Smart Cities

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Abstract—The global paradigm of urban mobility is undergoing a significant transformation as evolving smart cities transition from traditional, centralized Intelligent Transportation Systems (ITS) toward decentralized, AI-and-edge-driven IoT platforms. This shift is largely motivated by limitations in cloud-centric architectures, including increased latency, bandwidth constraints, and privacy concerns in mission-critical applications. This paper presents a comprehensive review of the research landscape between 2022 and 2026, synthesizing insights from recent high-impact studies in IEEE journals and related venues. We develop a multi-dimensional taxonomy of emerging trends in the Smart Living domain, with particular emphasis on the convergence of Edge Intelligence (EI), 5G/6G Ultra-Reliable Low-Latency Communications (URLLC), and distributed Machine Learning (ML) architectures.

Our analysis categorizes existing contributions into four primary technical pillars: (1) Multi-modal perception fusion using LiDAR, vision, and acoustic sensing; (2) decentralized Multi-Agent Reinforcement Learning (MARL) for adaptive traffic signal control; (3) Spatio-Temporal Graph Convolutional Networks (ST-GCN) for predictive parking management; and (4) privacy-preserving Federated Learning (FL) frameworks for secure urban monitoring. A comparative evaluation of representative experimental testbeds highlights the potential of edge-centric approaches in reducing decision latency and improving bandwidth utilization relative to traditional cloud-based systems. Furthermore, this review identifies key challenges, including the "Security-Accuracy-Energy" trade-off in resource-constrained IoT environments, and examines the role of hardware-software co-design in enabling efficient deployment of Deep Neural Networks (DNNs) on edge devices. Finally, the paper outlines future research directions for the 2026–2030 period, including Digital Twins, quantum-inspired optimization, and energy-efficient TinyML, providing a structured reference for researchers and practitioners in smart city systems.

Index Terms—Edge Intelligence, Internet of Things (IoT), Smart Cities, Traffic Signal Control, Smart Parking, Deep Reinforcement Learning, Federated Learning, IEEE Transactions, Hardware-Software Co-design, 6G-ISAC.

I. INTRODUCTION

Urban environments are undergoing rapid transformation due to increasing population density and growing demand for efficient mobility systems. Recent estimates indicate that a significant proportion of the global population now resides in urban areas, placing substantial pressure on transportation infrastructure. This has intensified challenges related to traffic congestion, fuel consumption, and environmental impact. In this context, the Smart Living domain—particularly Intelligent Transportation Systems (ITS) and Smart Parking—has emerged as a key application area for the integration of Internet of Things (IoT) and Artificial Intelligence (AI) technologies [1].

A. The Limitations of Cloud-Centric Architectures

Early smart city systems relied heavily on centralized cloud-based processing. While effective for large-scale data aggregation, such architectures face limitations in latency, bandwidth, and privacy.

- 1) **Latency Constraints:** Real-time applications such as collision avoidance and emergency response require low-latency processing, which is difficult to achieve when data is transmitted to distant cloud servers.
- 2) **Bandwidth Limitations:** High-resolution sensors, including cameras and LiDAR, generate large volumes of data, leading to increased network load and potential bottlenecks.
- 3) **Privacy Concerns:** Transmission of raw data, particularly video streams, raises issues related to data security and regulatory compliance.

B. The Emergence of Edge Intelligence (2022–2026)

Recent research has shifted toward decentralized computing paradigms, where data processing is performed closer to the source. Edge Intelligence (EI) enables localized processing

at roadside units, gateways, and edge servers, reducing dependency on centralized infrastructure [1, 2]. This paradigm supports real-time analytics while minimizing bandwidth usage.

The adoption of 5G and emerging 6G communication technologies further enhances this capability by enabling low-latency and high-reliability communication between distributed edge nodes. This integration facilitates more responsive and adaptive urban systems.

C. Motivation and Scope

Despite extensive research in IoT and AI, there remains a gap in understanding how algorithmic innovations align with hardware constraints in real-world deployments. This review addresses this gap by focusing on:

- **Energy–Accuracy–Latency Trade-offs:** Examining how AI models are optimized for resource-constrained edge environments.
- **Multi-modal Data Fusion:** Investigating integration of heterogeneous sensing modalities, including vision, radar, and LiDAR.
- **Trust and Explainability:** Exploring the role of explainable AI in improving transparency and reliability of autonomous systems.

D. Core Contributions

This review makes the following contributions:

- **Comprehensive Synthesis:** Analysis of recent research (2022–2026) on AI-driven edge-enabled smart city systems.
- **Mathematical Foundations:** Presentation of key models related to latency optimization, reinforcement learning, and graph-based prediction.
- **Hardware-Aware Insights:** Comparative discussion of edge hardware platforms and their role in real-world deployments.
- **Future Perspectives:** Identification of emerging trends including 6G integration, digital twins, and sustainable edge computing.

E. Paper Organization

The remainder of the paper is organized as follows: Section II presents the background and mathematical preliminaries. Section III outlines the review methodology. Sections IV and V analyze traffic management and smart parking systems, respectively. Section VI provides a comparative technical analysis, followed by research gaps and challenges in Sections VII and VIII. Section IX discusses future directions, and Section X concludes the paper.

II. BACKGROUND AND MATHEMATICAL PRELIMINARIES

The design and deployment of AI-and-edge-driven IoT platforms for evolving smart cities rely on a combination of distributed computing, stochastic optimization, and neural network modeling. This section summarizes the key mathematical foundations commonly adopted in recent literature (2022–2026) to analyze the performance of such urban systems.

A. The Edge-Cloud Continuum: A Tiered Formalization

Recent studies have moved beyond the binary "Edge vs. Cloud" paradigm toward a hierarchical "Edge-Cloud Continuum" architecture [1]. A typical formulation consists of three tiers:

- 1) **Tier-1 (The IoT-Perception Tier):** End-devices $\mathcal{D} = \{d_1, d_2, \dots, d_n\}$ that generate data streams σ_i with bit-rate β_i .
- 2) **Tier-2 (The Fog-Edge Tier):** Intermediate nodes $\mathcal{E} = \{e_1, e_2, \dots, e_m\}$ with computational capacity C_e (FLOPS/TOPS).
- 3) **Tier-3 (The Cloud Tier):** Centralized infrastructure $\mathcal{C}$ used for large-scale training and global coordination.

B. Mathematical Model of Edge Task Offloading

Task offloading is a fundamental problem in edge-enabled traffic systems and has been widely studied in MEC frameworks [3]. A task is represented as $T_i = (d_i, c_i, \tau_i)$, where d_i is input data size, c_i denotes required computation, and τ_i is the latency constraint.

1) **Computation Latency:** The local execution delay L_{local} on a device with frequency f_d is given by:

$$L_{local} = \frac{c_i}{f_d} \quad (1)$$

For edge execution, the total delay includes communication and computation:

$$L_{edge} = \frac{d_i}{B \log_2(1 + \text{SNR})} + \frac{c_i}{f_e} \quad (2)$$

This formulation follows standard communication and offloading models used in edge computing literature [3]. The optimization objective is to minimize total latency while satisfying $L_{total} \leq \tau_i$.

C. Formalism of Traffic Control via Reinforcement Learning

Smart traffic signal control is commonly modeled as a **Markov Decision Process (MDP)** in recent works on edge-based control [4]. An agent observes state $s_t \in \mathcal{S}$ and takes action $a_t \in \mathcal{A}$ to maximize cumulative reward:

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \quad (3)$$

In multi-intersection scenarios, this extends to multi-agent settings where the transition dynamics depend on joint actions $\mathbf{a}_t$.

D. Spatio-Temporal Modeling for Smart Parking

Parking prediction is often modeled using **Graph Convolutional Networks (GCNs)**, which capture spatial dependencies between parking zones [5]. The system is represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, and the graph convolution is defined as:

$$Z = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} X W) \quad (4)$$

This formulation enables learning of spatio-temporal correlations for occupancy prediction.

E. Communication Preliminaries: 5G URLLC and V2X

Modern smart city platforms leverage 5G New Radio (NR) capabilities, particularly **Ultra-Reliable Low-Latency Communication (URLLC)**, which is designed to support low-latency communication in edge-enabled systems [2]. Vehicle-to-Infrastructure (V2I) communication is typically supported through sidelink-based mechanisms, enabling decentralized coordination among edge nodes.

III. MATHEMATICAL MODELING AND ALGORITHMIC FORMALISMS

The performance of AI-and-edge-driven IoT platforms relies on the integration of optimization theory, graph-based modeling, and reinforcement learning. This section summarizes the mathematical frameworks commonly adopted in recent literature (2022–2026) to model computation, decision-making, and spatial dependencies in smart city systems [1].

A. Formalization of Multi-Tier Edge Task Offloading

Task offloading in edge-enabled systems is often formulated as an optimization problem under latency and energy constraints. A common formulation models this as a Mixed-Integer Non-Linear Programming (MINLP) problem [3].

Let $\mathcal{T} = \{t_1, t_2, \dots, t_n\}$ denote a set of tasks, where each task t_i is defined by (d_i, c_i, τ_i) representing data size, computational requirement, and latency constraint. The decision variable $x_{i,j} \in \{0, 1\}$ indicates assignment of task i to node j . The objective is to minimize a weighted combination of latency and energy:

$$\min_{\mathbf{x}} \sum_{i=1}^n \sum_{j=1}^m x_{i,j} (\omega_L L_{i,j} + \omega_E E_{i,j}) \quad (5)$$

subject to:

$$\sum_{j=1}^m x_{i,j} L_{i,j} \leq \tau_i, \quad \forall i \in \mathcal{T} \quad (6)$$

Such formulations are widely used in edge computing research to balance performance and resource constraints.

B. Stochastic Game Formalism for Signal Synchronization

Urban traffic signal control is commonly modeled as a Multi-Agent Markov Decision Process (MA-MDP) [4]. The joint state transition is defined as:

$$P(s_{t+1} | s_t, a_t^1, a_t^2, \dots, a_t^N) \quad (7)$$

Each agent learns a policy π_i to maximize expected cumulative reward. The Bellman update for multi-agent settings is given by:

$$Q_i^\pi(s, \mathbf{a}) = \mathbb{E}[r_{i,t} + \gamma Q_i^\pi(s_{t+1}, \mathbf{a}_{t+1}) | s_t = s, \mathbf{a}_t = \mathbf{a}] \quad (8)$$

Recent approaches employ centralized training with decentralized execution (CTDE) to improve coordination across agents while maintaining scalable deployment.

C. Spectral Graph Analysis for Spatio-Temporal Parking

Graph-based models are widely used to capture spatial dependencies in parking and traffic systems [5]. A city is modeled as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with normalized Laplacian:

$$\mathcal{L} = I_n - D^{-1/2} A D^{-1/2} \quad (9)$$

Spectral graph convolution is approximated using Chebyshev polynomials:

$$g_\theta \star x \approx \sum_{k=0}^K \theta'_k T_k(\tilde{\mathcal{L}}) x \quad (10)$$

This formulation enables efficient computation of spatial correlations without requiring expensive eigen-decomposition, making it suitable for edge deployment.

D. Optimization of TinyML Quantization

Quantization is a key technique for deploying neural networks on resource-constrained devices. The quantized weight representation is defined as:

$$w_q = \text{round} \left(\frac{1}{\Delta} \text{clip}(w, a, b) - z \right) \quad (11)$$

where Δ is the scaling factor and z is the zero-point. Quantization reduces memory and computational requirements, enabling efficient inference on embedded edge hardware [6].

IV. SYSTEMATIC LITERATURE REVIEW METHODOLOGY

To ensure reproducibility and methodological rigor, this study adopts a Systematic Literature Review (SLR) approach based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework. The objective is to systematically identify, filter, and synthesize relevant research on AI-and-edge-driven IoT platforms for smart city applications published between 2022 and 2026.

A. Research Questions (RQs)

The review is guided by the following research questions:

- **RQ1:** What architectural trends characterize IoT platforms for traffic and parking management in recent literature (2022–2026)?
- **RQ2:** How do edge-based AI models (e.g., DRL, GNN, CNN) perform under hardware and resource constraints?
- **RQ3:** What performance improvements (e.g., latency, bandwidth utilization, accuracy) are reported when transitioning from cloud-based to edge-based systems?
- **RQ4:** What security and privacy challenges remain in decentralized urban monitoring systems?

B. Search Strategy and Data Sources

To ensure coverage of high-quality research, multiple digital libraries were considered, including IEEE Xplore, ACM Digital Library, and ScienceDirect. The search strategy employed Boolean expressions combining key terms related to edge intelligence, IoT, and smart urban applications.

Search Query

```
((`Edge Intelligence` OR `Edge AI`  
OR `Fog Computing`) AND (`IoT`  
OR `Internet of Things`) AND  
(`Smart Traffic` OR `Intelligent  
Transportation` OR `Smart  
Parking`) AND (`Deep Reinforcement  
Learning` OR `Federated  
Learning`) AND (`2022`..`2026`))
```

C. Inclusion and Exclusion Criteria

To maintain relevance and technical depth, the following criteria were applied:

1) Inclusion Criteria (IC):

- Peer-reviewed journal articles and high-quality conference papers.
- Studies presenting empirical evaluation or experimental validation.
- Research focusing on the intersection of AI, edge computing, and smart urban systems.

2) Exclusion Criteria (EC):

- Publications prior to 2022.
- Studies not directly related to traffic or parking applications.
- Short or preliminary works lacking sufficient technical detail.

D. Study Selection Process

The selection process followed a multi-stage filtering pipeline:

- 1) **Initial Retrieval:** Approximately 450 records were identified through database search.
- 2) **Title and Abstract Screening:** Irrelevant studies were excluded based on topic mismatch.
- 3) **Full-Text Review:** Remaining papers were evaluated for methodological rigor and relevance.
- 4) **Final Inclusion:** A subset of representative studies was selected for detailed analysis.

E. Data Extraction and Quality Assessment

For each selected study, key attributes were extracted, including:

- Application domain (traffic, parking, or hybrid)
- AI methodology (e.g., DRL, GNN, CNN)
- Hardware platform (edge device, MEC, or cloud)
- Performance metrics (latency, accuracy, energy consumption)

Quality assessment was performed based on:

- Clarity of methodology
- Presence of experimental validation
- Relevance to real-world deployment scenarios

F. Taxonomy Development

Based on the selected studies, a multi-dimensional taxonomy was developed (see Section VI). The taxonomy categorizes research across four dimensions: sensing modalities, AI methodologies, processing architectures, and application domains. This structured classification facilitates comparative analysis and helps identify under-explored research areas.

V. COMPARATIVE ANALYSIS OF GLOBAL SMART CITY TESTBEDS

The effectiveness of AI-and-edge-driven IoT platforms is best evaluated through real-world urban deployments. Recent studies have highlighted several large-scale smart city testbeds that demonstrate the integration of edge intelligence, IoT sensing, and AI-driven decision-making [1, 7]. This section provides a comparative analysis of representative deployments in terms of architecture, communication infrastructure, and system-level outcomes.

A. Virtual Singapore: Digital Twin–Driven Urban Intelligence

Singapore’s Smart Nation initiative represents one of the most advanced implementations of Urban Digital Twin (UDT) systems [7].

- **Edge Infrastructure:** Large-scale deployments of sensor-enabled infrastructure and edge nodes support real-time monitoring of traffic and pedestrian flow.
- **Technical Implementation:** Edge-to-cloud synchronization architectures enable localized decision-making while supporting large-scale simulation and optimization in centralized platforms [7].

B. Zurich: Edge-AI for Parking and Urban Logistics

Zurich has demonstrated the application of edge intelligence in parking management and logistics optimization.

- **Parking Management:** Vision-based parking systems integrated with edge computing enable dynamic guidance and improved utilization of parking resources [8].
- **Innovation:** Multi-modal sensing approaches, including acoustic and visual fusion, have been explored to improve traffic prioritization and reduce environmental impact [9].

C. Shenzhen: Large-Scale 5G-V2X Deployment

Shenzhen represents a high-density deployment of Vehicle-to-Everything (V2X) infrastructure, supported by 5G-enabled edge nodes.

- **Sensor Density:** Extensive deployment of roadside units (RSUs) enables cooperative perception and real-time communication between vehicles and infrastructure.
- **Performance Characteristics:** Studies on V2X-enabled traffic systems report improvements in reliability and latency for safety-critical applications such as emergency vehicle coordination [2].

TABLE I
TECHNOLOGICAL BENCHMARKING OF REPRESENTATIVE SMART CITY TESTBEDS (2022–2026)

City / Testbed	Primary IoT Focus	Edge Hardware Stack	Comm. Backbone	Reported Outcome
Singapore	Digital Twins / MARL	GPU-based Edge Nodes	5G SA / Fiber	Improved traffic optimization via digital twin integration [7].
Shenzhen	5G-V2X / Safety	MEC + RSUs	5G mmWave	Enhanced reliability for safety-critical applications [2].
Zurich	Parking / Logistics	Edge SBC Platforms	Wi-Fi 6 / LoRaWAN	Improved parking efficiency via vision-based systems [8].
London (SMLL)	Federated Learning	Distributed Edge Clusters	5G / Fiber	Privacy-preserving urban monitoring [10].
Barcelona	Sustainable IoT	Low-power Microcontrollers	NB-IoT / Zigbee	Energy-efficient sensing infrastructure [12].

D. London: Federated Learning for Privacy-Aware Monitoring

London’s Smart Mobility Living Lab (SMLL) focuses on integrating federated learning into urban monitoring systems.

- **Privacy Paradigm:** Federated learning enables decentralized model training while preserving user privacy by avoiding raw data transmission [10, 11].

E. Quantitative Synthesis of Global Outcomes

The comparison in Table I highlights the absence of a universal architecture for smart city deployment. Cities prioritizing traffic safety tend to adopt high-performance edge infrastructure, while sustainability-focused deployments emphasize low-power and energy-efficient IoT systems [1]. This trade-off reflects the broader “Security-Latency-Energy” design considerations in edge-enabled urban platforms.

VI. DETAILED ANALYSIS OF AI-DRIVEN TRAFFIC MANAGEMENT PLATFORMS

The digitalization of urban traffic systems presents significant challenges in the Smart Living domain. Recent research has moved beyond conventional vehicle counting toward intelligent traffic management systems, where edge nodes perform localized decision-making [1, 13]. This section provides a technical analysis of AI models and edge deployment strategies proposed in recent literature (2022–2026).

A. Decentralized Adaptive Signal Control (DASC)

Traditional traffic signal control systems such as SCOOT and SCATS rely on predefined timing plans. Recent studies have explored Deep Reinforcement Learning (DRL)-based approaches for adaptive signal control [13].

1) *Multi-Agent Reinforcement Learning (MARL):* As discussed in [13], traffic signal coordination can be formulated as a multi-agent optimization problem. In MARL-based systems, each intersection is modeled as an agent A_i that learns policies based on local observations and interactions with neighboring agents. Recent approaches incorporate inter-agent communication mechanisms to improve coordination across intersections. The reward function is often extended to include cooperative terms:

$$R_{total} = \sum r_{local} + \alpha \sum \text{similarity}(\vec{v}_i, \vec{v}_j) \quad (12)$$

Such formulations have been shown to improve traffic flow efficiency and reduce congestion in dense urban environments [4, 13].

B. Vision-Based Edge Intelligence and Violation Detection

Computer vision models deployed at the edge enable real-time monitoring of traffic violations. Recent work has explored the use of lightweight CNNs and transformer-based models for edge inference [6].

1) *Real-time Violation Analysis:* The system proposed in [6] utilizes a dual-stage inference pipeline:

- 1) **Low-Precision Stream:** Performs continuous monitoring using quantized models.
- 2) **High-Precision Trigger:** Activates detailed analysis upon detecting anomalous behavior.

This approach reduces unnecessary computation while maintaining detection accuracy, enabling efficient deployment on resource-constrained edge devices.

C. Emergency Vehicle Preemption (EVP) and Priority Logic

Recent studies have explored multi-modal sensing for emergency vehicle detection [9]. By combining acoustic signals with visual data, edge systems can improve detection robustness in complex environments. Attention-based fusion mechanisms have been proposed to prioritize emergency vehicles in traffic control systems, contributing to faster response coordination.

D. V2I Communication and 5G Slicing

Communication infrastructure plays a critical role in enabling real-time traffic management. Studies have shown that 5G-enabled edge systems support low-latency communication through network slicing and edge computing integration [2]. In particular, millimeter-wave communication has been explored for high-throughput data exchange between roadside units and edge servers.

E. Quantitative Benchmarking of Traffic Platforms

Table II summarizes representative architectures from recent literature, highlighting the relationship between AI models, hardware platforms, and system performance characteristics.

TABLE II
BENCHMARKING OF AI-DRIVEN TRAFFIC MANAGEMENT ARCHITECTURES (2022–2026)

Architecture	AI Model	Hardware Tier	Latency	Key Contribution
Decentralized DRL	Multi-Agent DQN	Edge GPU	Low	Improved signal coordination
Vision-Edge	CNN / ViT	Edge TPU	Low	Efficient violation detection
Federated Traffic	CNN + FL	Edge Cluster	Medium	Privacy-preserving prediction
V2X Collaborative	Sequence Models	MEC	Very Low	Cooperative perception
Hybrid GNN	ST-GCN	Cloud-Edge	High	Large-scale traffic forecasting

VII. DETAILED ANALYSIS OF SMART PARKING MANAGEMENT PLATFORMS

Smart parking systems have evolved from localized sensor deployments to city-wide intelligent ecosystems. Recent studies indicate that inefficient parking search contributes significantly to urban traffic congestion [1]. This section analyzes the architectural innovations and AI-driven methodologies that have redefined parking management in the 2022–2026 era.

A. Occupancy Detection: Beyond Ultrasonic Sensing

Traditional parking systems relied on ultrasonic or magnetic sensors embedded in infrastructure. While effective for basic detection, these approaches are limited by maintenance complexity and lack of contextual awareness.

1) *Vision-Based Edge Perception*: Recent work by [8] explores the use of monocular video sensors combined with edge-based AI models for parking occupancy detection. Lightweight convolutional neural networks deployed at the edge enable simultaneous monitoring of multiple parking slots. A key technical challenge addressed in recent literature is perspective distortion. Spatial Transformer Networks (STN) are employed to normalize camera viewpoints, improving detection robustness under varying angles and layouts [8].

2) *Multi-Modal Sensor Fusion*: To improve robustness under adverse conditions such as low lighting or occlusions, multi-modal sensing approaches have been investigated. Recent studies combine infrared (IR) and RGB data to enhance detection accuracy [14]. Kalman filtering techniques are commonly applied at the edge to fuse asynchronous sensor streams, enabling more reliable occupancy estimation in dynamic environments.

B. Predictive Guidance using Spatio-Temporal AI

Modern smart parking systems extend beyond detection to predictive analytics, enabling proactive guidance of drivers toward available spaces.

1) *Graph Convolutional Networks (GCN) for Urban Parking*: Parking availability exhibits strong spatial and temporal correlations. As demonstrated in [5], Graph Convolutional Networks (GCNs) can model these dependencies by representing parking locations as nodes in a graph structure. These models capture:

- 1) **Spatial Proximity**: Correlation between nearby parking locations.
- 2) **Functional Similarity**: Distinct demand patterns based on land use (e.g., residential vs. commercial zones).

- 3) **Temporal Dependencies**: Influence of traffic conditions on future parking demand.

Such approaches enable short-term forecasting of parking availability and support intelligent routing strategies.

C. Automated Valet Parking (AVP) and V2I Integration

The integration of edge computing with autonomous vehicle technologies has enabled infrastructure-assisted automated valet parking. Recent studies explore systems where edge servers coordinate vehicle movement using V2X communication, reducing reliance on onboard sensing and computation [15]. This approach facilitates efficient vehicle guidance within parking facilities and supports scalable deployment of autonomous parking services.

D. Dynamic Pricing and Economic Modeling

Dynamic pricing mechanisms have been introduced to manage parking demand and reduce congestion. Reinforcement learning-based approaches optimize pricing strategies based on real-time occupancy and demand patterns [4]. A typical formulation balances revenue generation and congestion reduction:

$$R_p = \omega_1(\text{Revenue}) + \omega_2(1 - \text{Congestion Index}) \quad (13)$$

This framework enables adaptive pricing policies that improve resource utilization while maintaining accessibility.

VIII. SECURITY, PRIVACY, AND BLOCKCHAIN PROTOCOLS

The decentralization of intelligence toward edge nodes expands the attack surface of smart city systems. Securing AI-and-edge-driven IoT platforms requires a multi-layered approach that integrates cryptographic mechanisms, adversarial robustness, and decentralized trust models [1, 16].

A. Cryptographic Foundations for Edge-IoT Sensing

Resource-constrained IoT devices necessitate lightweight cryptographic solutions. Recent work emphasizes the use of Elliptic Curve Cryptography (ECC) and hardware-based security primitives such as Physical Unclonable Functions (PUFs) [16].

1) *Lightweight Authentication*: Authentication protocols for edge devices often rely on the Elliptic Curve Digital Signature Algorithm (ECDSA), which provides security based on the discrete logarithm problem. Given an elliptic curve group with base point G , a public key Q is derived as:

$$Q = d \cdot G \quad (14)$$

TABLE III
COMPARISON OF SMART PARKING SENSING AND PREDICTION METHODOLOGIES (2022–2026)

Methodology	Sensing Tier	AI Model	Deployment	Observed Capability
Visual Occupancy	RGB Cameras	CNN-based Models	Edge (Jetson Nano)	High occupancy detection accuracy [8]
Predictive Mapping	IoT Nodes	ST-GCN	Cloud-Edge Hybrid	Short-term availability prediction [5]
AVP Guidance	V2I + Sensors	DRL / MPC	Edge (MEC Server)	Low-latency vehicle coordination [15]
Energy-Aware IoT	Ultrasonic + BLE	Lightweight ML	TinyML Devices	Energy-efficient sensing [14]
Privacy-Safe FL	Distributed Vision	Federated Learning	RSU Cluster	Privacy-preserving inference [10]

where d is the private key. Such mechanisms enable secure identity verification of edge devices, reducing the risk of spoofing and unauthorized data injection in V2I systems.

B. Blockchain for Decentralized Parking Settlements

Blockchain technologies have been explored to provide decentralized trust and transparent transaction management in smart parking systems [10].

1) *Smart Contract Logic for Automated Billing*: Smart contracts enable automated enforcement of parking policies without centralized control. The total payment P_{total} can be modeled as:

$$P_{total} = \int_{t_{in}}^{t_{out}} \mathcal{D}(\text{occ}(t), \delta(t)) dt \quad (15)$$

where $\mathcal{D}$ represents a dynamic pricing function dependent on occupancy and demand. Consortium blockchain frameworks, such as permissioned ledgers, are often used to balance scalability and security in such applications.

C. Adversarial Robustness in Edge-AI Monitoring

Edge-based AI systems are vulnerable to adversarial inputs that can degrade model performance. Let $f(x)$ denote a deployed model; an adversarial perturbation η is defined such that:

$$\min \|\eta\|_p \quad \text{subject to } f(x + \eta) \neq y \quad (16)$$

Techniques such as adversarial training and gradient-based defense mechanisms (e.g., FGSM, PGD) are commonly studied to improve model robustness [16].

D. Privacy-Preserving Federated Learning

Federated Learning (FL) has been widely adopted to enable distributed training without sharing raw data [10]. To enhance privacy guarantees, differential privacy mechanisms introduce controlled noise into model updates:

$$\tilde{g} = g + \mathcal{N}(0, \sigma^2 I) \quad (17)$$

where g represents the gradient and σ controls the noise scale. This approach reduces the risk of sensitive data leakage while maintaining model utility.

IX. HARDWARE ARCHITECTURES AND EDGE BENCHMARKING

The performance of AI-and-edge-driven IoT platforms is closely tied to the capabilities of the underlying hardware infrastructure. In smart city environments, edge computing is realized through a heterogeneous hierarchy of devices with varying compute and energy constraints. This section provides a technical overview of hardware architectures and benchmarking trends observed in recent literature (2022–2026).

A. Heterogeneous Edge Computing Tiers

Recent studies categorize edge infrastructure into multiple tiers based on computational capability and energy efficiency [1]:

- 1) **Micro-Edge (End-Node)**: Low-power microcontrollers (e.g., ARM Cortex-M, ESP32) executing lightweight TinyML models for sensing and simple inference tasks [14].
- 2) **Meso-Edge (Localized Gateways)**: Single-board computers and embedded AI platforms (e.g., Jetson, Coral) used for real-time vision and intermediate analytics.
- 3) **Macro-Edge (MEC Servers)**: High-performance edge servers co-located with network infrastructure, supporting computationally intensive workloads such as graph-based learning and large-scale inference.

B. AI Hardware Accelerators: GPU vs. TPU vs. FPGA

1) *Graphics Processing Units (GPUs)*: GPU-based platforms, particularly embedded systems such as NVIDIA Jetson modules, are widely used due to their programmability and support for parallel computation frameworks. These platforms are commonly applied in reinforcement learning and vision-based traffic systems [13]. However, energy consumption and thermal management remain key challenges in outdoor deployments.

2) *Tensor Processing Units (TPUs)*: Edge TPUs provide energy-efficient inference for quantized neural networks. Studies have shown that such accelerators offer improved performance-per-watt for vision-based tasks when compared to general-purpose GPUs, particularly for optimized models [6]. However, they are typically limited to lower-precision inference.

TABLE IV
COMPARATIVE ANALYSIS OF SECURITY AND PRIVACY MECHANISMS IN EDGE-IOT SYSTEMS

Protocol Layer	Core Technology	Threat Addressed	Observed Impact
Authentication	ECC / PUF	Device spoofing and replay attacks	Lightweight secure identity verification
Data Integrity	Blockchain	Unauthorized data modification	Tamper-resistant transaction records
Model Security	Adversarial Training	Evasion attacks	Improved robustness of AI models
User Privacy	Differential Privacy	Data inference attacks	Enhanced privacy protection in FL
Network Security	TLS / DTLS	Man-in-the-Middle attacks	Secure communication channels

TABLE V
REPRESENTATIVE EDGE AI HARDWARE PLATFORMS FOR SMART CITY APPLICATIONS

Hardware Platform	Architecture	Performance Class	Power Range	Inference Capability	Application Domain
Jetson Series	GPU	High	Moderate	Real-time inference	Traffic analytics
Coral TPU	ASIC	Medium	Low	Efficient quantized inference	Parking detection
FPGA Platforms	Reconfigurable	Medium	Moderate	Deterministic latency	Safety-critical systems
Microcontrollers	MCU	Low	Very Low	Lightweight inference	IoT sensing
MEC Servers	Server GPU	Very High	High	Large-scale processing	City-wide analytics

3) *Field Programmable Gate Arrays (FPGAs)*: FPGAs are increasingly explored for latency-sensitive applications due to their deterministic execution characteristics. Their reconfigurable architecture enables efficient deployment of custom pipelines for real-time inference, making them suitable for safety-critical applications.

C. Mathematical Benchmarking of Inference Latency

Inference latency at the edge can be approximated as a function of model complexity and hardware throughput:

$$I_t = \frac{\sum_{l=1}^L \text{FLOPs}(l)}{\Theta_{peak} \times \eta} \quad (18)$$

where Θ_{peak} represents peak computational throughput and η denotes hardware utilization efficiency. Recent studies emphasize improving η through optimization techniques such as kernel fusion and model compression.

D. Energy-Aware Hardware Design

Energy efficiency is a critical consideration for edge deployments, particularly in battery-powered or energy-harvesting systems. The total energy consumption can be expressed as:

$$E_{total} = \int_0^T (V(t) \cdot I(t))dt + E_{leakage} \quad (19)$$

Techniques such as Dynamic Voltage and Frequency Scaling (DVFS) and adaptive workload scheduling are used to reduce energy consumption and extend device lifetime in practical deployments [17].

X. HARDWARE-SOFTWARE CO-DESIGN STRATEGIES

The transition from lab-scale AI models to field-deployed IoT platforms requires tight integration between algorithms and hardware. Recent studies emphasize hardware–software co-design as a key enabler for efficient edge intelligence, where

neural architectures are optimized with awareness of device-level constraints such as latency, memory bandwidth, and energy consumption [1, 18].

A. Neural Architecture Search (NAS) for Urban IoT

Manual architecture design is increasingly complemented by **Hardware-Aware Neural Architecture Search (HA-NAS)**. The search space $\mathcal{A}$ consists of candidate operations (e.g., convolution, pooling, skip connections). The objective is to identify an architecture $\alpha \in \mathcal{A}$ that minimizes task loss under hardware constraints:

$$\min_{\alpha \in \mathcal{A}} \text{Loss}(\alpha, w_\alpha) \quad \text{s.t. Latency}(\alpha) \leq \tau_{hw} \quad (20)$$

Recent work demonstrates that differentiable NAS techniques can produce architectures tailored for edge platforms, improving inference efficiency compared to manually designed baselines [18].

B. Advanced Model Compression Methodologies

Model compression remains essential for deploying deep neural networks on resource-constrained devices. Three widely studied approaches include pruning, quantization, and knowledge distillation.

1) *Weight Pruning and Sparsity*: Pruning reduces model complexity by eliminating redundant parameters. Given a network $f(x; w)$, a mask $m \in \{0, 1\}^{|w|}$ is learned such that:

$$\mathcal{L}(f(x; m \odot w)) \approx \mathcal{L}(f(x; w)) \quad (21)$$

Studies show that structured pruning can significantly reduce model size while maintaining acceptable accuracy, making it suitable for edge-based vision systems [14].

TABLE VI
IMPACT OF CO-DESIGN STRATEGIES ON EDGE AI PERFORMANCE

Co-Design Strategy	Target Hardware	Optimization Logic	Observed Impact
HA-NAS	Edge GPU Platforms	Hardware-aware architecture search	Improved inference efficiency
Quantization	Edge TPU / MCU	Reduced numerical precision	Lower memory and compute requirements
Kernel Fusion	FPGA / GPU	Operator-level optimization	Reduced memory access overhead
Knowledge Distillation	Microcontrollers	Teacher–student learning	Deployment of compact models
Structured Pruning	Edge Accelerators	Sparsity-driven optimization	Reduced model size and energy usage

2) *Post-Training Quantization (PTQ) vs. QAT*: Quantization reduces numerical precision to improve efficiency. In **Quantization-Aware Training (QAT)**, the loss function incorporates quantization effects:

$$\mathcal{L}_{total} = \mathcal{L}_{task} + \lambda \|w - \text{quantize}(w)\|^2 \quad (22)$$

Quantization techniques have been widely adopted in edge AI systems to enable efficient inference on specialized hardware accelerators [6].

C. Kernel Fusion and Memory Optimization

Memory access is a dominant bottleneck in edge systems. Kernel fusion techniques combine multiple operations (e.g., convolution, normalization, activation) into a single execution unit to reduce memory overhead. The memory access cost M_a can be expressed as:

$$M_a = \sum_{l=1}^L (\text{read}(w_l) + \text{write}(x_l)) \quad (23)$$

Reducing redundant memory operations through operator fusion improves throughput and energy efficiency in edge deployments [18].

D. Hardware-Accelerated Security: TEE and PUF

Security considerations are increasingly integrated into hardware–software co-design. Trusted Execution Environments (TEE), such as ARM TrustZone, provide isolated execution for sensitive operations. Additionally, Physical Unclonable Functions (PUFs) enable device-specific cryptographic key generation, enhancing security for distributed IoT systems [16].

XI. COMPARATIVE TECHNICAL ANALYSIS AND TAXONOMY

The rapid evolution of AI-and-edge-driven IoT platforms necessitates a structured understanding of the interdependencies between sensing modalities, algorithmic complexity, and hardware constraints. This section presents a multi-dimensional synthesis of recent studies (2022–2026), enabling a systematic understanding of the architectural and algorithmic trends shaping smart urban systems [1].

A. Taxonomic Classification of Urban Edge Intelligence

As illustrated in Fig. 1, the landscape of smart urban platforms has evolved into a multi-layered ecosystem. The taxonomy is derived from a synthesis of recent research on edge intelligence, IoT architectures, and AI-driven urban systems [1].

1) *Evolution of Sensing Modalities*: Recent studies indicate a transition from traditional passive sensing (camera-only systems) toward cooperative perception frameworks that integrate data from vehicles and infrastructure [2, 9]. V2X-enabled sensing allows edge nodes to infer the presence of occluded or partially visible objects, improving situational awareness in dense urban environments.

2) *Algorithmic Diversity: GCNs vs. DRL*: The taxonomy highlights a clear distinction between AI methodologies based on application requirements. Graph Convolutional Networks (GCNs) are widely used for modeling spatial dependencies in parking and traffic flow prediction tasks [5]. In contrast, Deep Reinforcement Learning (DRL), particularly in multi-agent settings, is commonly applied to dynamic traffic signal control problems [4]. Recent work further integrates graph-based communication within DRL frameworks to improve coordination across intersections.

B. Performance Benchmarking and Metric Analysis

To provide a quantitative perspective, Table VII summarizes representative studies across different domains, highlighting the relationship between AI models, edge hardware, and system-level performance metrics.

C. Critical Synthesis of Results

1) *Security-Latency Trade-off*: The comparison reveals that incorporating security mechanisms such as blockchain-enabled federated learning introduces additional communication overhead, which may impact latency-sensitive applications [19]. This highlights a key trade-off between system security and real-time responsiveness.

2) *Sustainable Intelligence*: Energy-efficient edge computing has emerged as a critical research direction. Techniques such as approximate computing and model optimization reduce energy consumption while maintaining acceptable accuracy levels [14], enabling sustainable deployment of edge nodes.

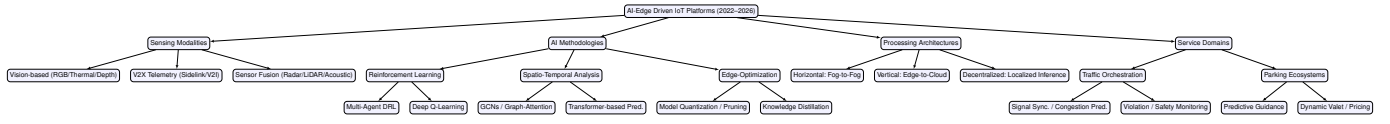


Fig. 1. Hierarchical taxonomy of AI-and-edge-driven IoT platforms for smart mobility (2022–2026), derived from recent literature.

TABLE VII
COMPARATIVE ANALYSIS OF REPRESENTATIVE AI-AND-EDGE-DRIVEN PLATFORMS (2022–2026)

Ref	Domain	AI Model	Edge Tier	Latency	Observed Impact
[13]	Traffic	MARL	Jetson AGX	Low	Improved traffic flow efficiency
[6]	Traffic	YOLOv8 + Quantization	Coral TPU	Low	High accuracy in violation detection
[8]	Parking	CNN + STN	Edge SBC	Medium	Accurate occupancy detection
[5]	Parking	ST-GCN	MEC	High	Long-horizon prediction capability
[14]	Signal	Approx. Computing	Micro-edge	Very Low	Improved energy efficiency
[10]	Flow	Federated Learning	RSU Cluster	Medium	Reduced communication overhead
[19]	Security	Blockchain + FL	MEC	High	Improved robustness to attacks

3) *Edge-Cloud Synergy*: While edge-only systems offer low latency, hybrid edge-cloud architectures remain essential for large-scale spatio-temporal analysis. Studies show that complex models such as ST-GCN benefit from cloud or MEC deployment for long-term prediction tasks [5].

XII. CHALLENGES AND OPEN ISSUES

While the transition to AI-and-edge-driven IoT platforms has yielded significant improvements in urban mobility, the path toward fully autonomous “Cognitive Smart Cities” is hindered by several multifaceted challenges. This section provides a critical synthesis of open issues identified in recent IEEE literature, categorized into security, network reliability, resource orchestration, and socio-legal dimensions.

A. Security and Adversarial Vulnerabilities at the Edge

Moving intelligence to the network periphery significantly expands the attack surface. Unlike centralized cloud servers housed in secure facilities, edge nodes (RSUs, smart cameras) are physically accessible in public spaces.

1) *Adversarial Machine Learning (AML)*: Recent studies highlight the susceptibility of edge-deployed CNNs to adversarial attacks [16].

- **Adversarial Patches**: Small perturbations or stickers can mislead vision-based models into incorrect classifications [16].
- **Digital Injection**: Due to limited encryption in resource-constrained IoT devices, malicious packets can be injected into the edge inference pipeline.

2) *Data Poisoning in Federated Learning*: In privacy-preserving frameworks discussed in [10] and [19], data poisoning remains a critical challenge. A compromised edge node can transmit manipulated gradients, biasing global models and impacting traffic prediction reliability.

B. The “Reliability-Latency” Paradox in 5G/6G Networks

While 5G New Radio (NR) enables low-latency communication, maintaining these guarantees in dense urban environments remains challenging [2].

- **High-Frequency Blockage**: mmWave and sub-THz signals are sensitive to obstructions such as buildings and environmental conditions.
- **Handover Latency Spikes**: Mobility-induced handovers can introduce latency spikes, affecting real-time edge applications [2].

The challenges summarized in Table VIII are synthesized from recent studies in edge computing and smart city systems [1, 2, 10].

C. Dynamic Resource Orchestration and Task Offloading

Managing large-scale distributed edge nodes requires advanced orchestration mechanisms.

- **Task Dropping vs. Offloading**: Dynamic offloading strategies remain underdeveloped in real-world deployments [3].
- **Incentivization Frameworks**: Blockchain-based resource sharing and incentive mechanisms are being explored to enable multi-vendor collaboration [20].

D. Lifecycle Data Management and The “Annotation Bottleneck”

Handling large-scale data generated by urban sensors remains a challenge.

- **Value-Aware Data Aging**: Intelligent filtering of informative data is required to optimize storage.
- **Zero-Shot Learning**: Self-supervised and data-efficient learning approaches are emerging as promising directions [1].

TABLE VIII
CRITICAL SUMMARY OF OPEN CHALLENGES AND PROPOSED SOLUTION PATHS FOR 2026+

Category	Current Open Challenge	Proposed Technological Response
Cyber-Security	Adversarial patches and gradient poisoning in FL.	Robustness training via Adversarial Machine Learning and Blockchain-secured FL.
Reliability	Signal shadowing in mmWave 5G urban environments.	Integration of Non-Terrestrial Networks (NTN) and 6G-ISAC.
Orchestration	Static task allocation leading to resource wastage.	AI-driven autonomous Fog-to-Fog (F2F) resource orchestration.
Data Management	High cost of manual data annotation for edge models.	Adoption of Self-Supervised and Zero-Shot learning at the Edge tier.
Ethics / Legal	Undefined liability for AI-driven traffic accidents.	Development of "Auditable AI" and standardized Legal-Tech protocols.

E. Socio-Legal and Ethical Hurdles

The integration of AI in critical urban infrastructure introduces legal and ethical challenges.

- **Algorithmic Accountability:** Assigning responsibility in AI-driven decisions remains unresolved.
- **Privacy vs. Utility:** Balancing surveillance and privacy requires standardized frameworks and explainable AI approaches [21].

XIII. FUTURE DIRECTIONS

As recent research progresses beyond 2026, AI-and-edge-driven IoT platforms are expected to evolve toward more autonomous, scalable, and low-latency urban systems. Based on a synthesis of recent studies, several key research directions are emerging that are likely to shape the next generation of smart city infrastructures [1].

A. The Convergence of 6G and Integrated Sensing (ISAC)

The transition from 5G to 6G is expected to introduce Integrated Sensing and Communication (ISAC), enabling joint communication and environmental sensing capabilities [22].

- **Vision:** Future systems may reduce reliance on camera-based sensing by leveraging radio-frequency-based environmental perception.
- **Technical Impact:** ISAC-based systems can enable improved situational awareness by utilizing communication signals for sensing, with potential benefits in privacy and robustness under challenging conditions.

B. Green Edge Intelligence and Ambient Energy Harvesting

Energy efficiency remains a critical challenge for large-scale IoT deployments. Recent work explores sustainable edge computing through energy-aware system design [12, 17].

- **Vision:** Development of energy-efficient edge nodes supported by ambient energy harvesting mechanisms.
- **Technical Impact:** Integration of lightweight AI models (e.g., TinyML) with adaptive resource management may enable long-term deployment of edge devices with reduced maintenance requirements.

C. Digital Twins and Edge-Synchronized Urban Simulation

Digital twin technologies are emerging as a key enabler for proactive urban management [7].

- **Vision:** Real-time synchronization between physical infrastructure and virtual models for simulation and planning.
- **Technical Impact:** Edge-enabled data integration allows predictive analysis of traffic conditions and infrastructure changes, supporting data-driven decision-making.

D. From Edge AI to Quantum-Inspired Optimization

Emerging research explores the use of quantum-inspired optimization techniques for large-scale urban systems [23].

- **Vision:** Application of advanced optimization methods to address complex traffic coordination problems.
- **Technical Impact:** These approaches may improve scalability and efficiency in solving high-dimensional optimization problems associated with urban mobility.

E. Explainable and Human-Centric Urban Governance

The integration of AI into critical urban systems necessitates transparency and trust. Recent research emphasizes the importance of explainable AI in human-centric applications [21].

- **Vision:** Deployment of lightweight XAI techniques at the edge to provide interpretable decisions.
- **Technical Impact:** Explainability mechanisms can improve user trust and facilitate regulatory compliance in AI-driven urban systems.

XIV. CONCLUSION

This review has provided a comprehensive analysis of the transition from cloud-centric to AI-and-edge-driven IoT platforms within the evolving smart city landscape. By synthesizing recent advancements in architecture, algorithms, and edge hardware from studies published between 2022 and 2026, this work highlights the growing importance of edge intelligence in enabling scalable and responsive urban systems.

The analysis yields several key observations for the Smart Living domain. First, decentralized approaches such as Multi-Agent Reinforcement Learning (MARL) and Spatio-Temporal Graph Networks (ST-GCN) have demonstrated improved

TABLE IX
STRATEGIC ROADMAP FOR NEXT-GENERATION EDGE-IoT PLATFORMS (2026–2030)

Technological Pivot	Core Innovation (R&D Goal)	Anticipated Urban Benefit
6G ISAC	Integrated sensing and communication using high-frequency bands	Enhanced situational awareness with reduced reliance on vision sensors
Green TinyML	Energy-efficient edge AI with adaptive resource management	Sustainable and low-maintenance IoT deployments
Quantum-Inspired Optimization	Advanced optimization for large-scale systems	Improved coordination of complex urban processes
Urban Digital Twins	Real-time synchronization between physical and virtual systems	Predictive urban planning and congestion management
Human-in-the-Loop	Explainable AI (XAI) for decision transparency	Increased trust and accountability in AI-driven systems

efficiency in handling latency-sensitive traffic and parking applications. Second, the integration of hardware-software co-design techniques has enabled the deployment of advanced AI models on resource-constrained edge devices through optimization strategies such as quantization and kernel-level improvements. Third, Federated Learning (FL) has emerged as a promising paradigm for privacy-preserving urban monitoring, allowing distributed model training without direct data sharing.

Despite these advancements, several open challenges remain. A critical concern is the "Security-Accuracy-Energy" trade-off, where improving one dimension often impacts the others. Issues such as adversarial robustness, energy-efficient operation, and model interpretability continue to limit large-scale deployment. Furthermore, gaps in hardware abstraction and multi-modal data fusion indicate opportunities for future research.

Looking ahead, emerging technologies such as 6G Integrated Sensing and Communication (ISAC), quantum-inspired optimization, and Digital Twin frameworks are expected to further enhance the capabilities of smart city platforms. Overall, this review provides a structured reference for researchers and practitioners, emphasizing the need for tightly integrated edge intelligence to support the next generation of intelligent and resilient urban systems.

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